SUBJECT: QBA 2

DATE: / /

* Data: Raw numbers and facts.
  - → processing → informations → making decision.

steps:

① data collection:

1. purpose of data.
2. what data is needed { qualitative. / quantitative.
3. design the data collection.
   - → amount of data.
     - → source of data.
       - → means of collection.
4. actually collecting data.

② Processing → information.

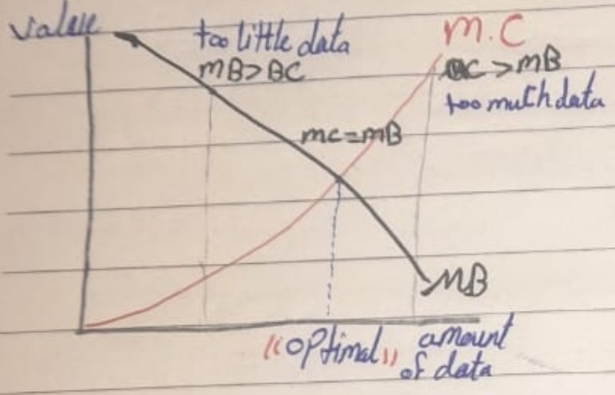
③ presentation.

SUBJECT: _____ DATE: / /

*amount of data:

⟹marginal cost: The extra cost of collecting one more bit of data.

⟹marginal benefit: Benefit of collecting one more bit of data.



Types of data: { qualitative / quantitative

- Nominal
→That type of data that cannot quantify with any meaningful units.

- ordinal
(Rank lorder)
→it's one more step quantify than nominal.
→it can rank categories of observation into some meaning order or rank.

- cardinal
(quantify)
*discrete: only integer numbers.
*Continous: can take an value and its not ristric to integer.

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Break even Point

- benefit = zero
- revenue: price/unit X No. unit sold.
- expense: Fixed cost + Variable cost.
  = Fixed cost + (variable cost/unit X No. unit sold)
- profit = Revenue - expenses

$$SX - (F + VX)$$

$$P = SX - F - VX$$

* S → selling price
* X → No. unit sold
* V → variable/unit
* F → fixed cost

$$\rightarrow 0 = SX - F - VX$$

$$0 = SX - VX - F$$

$$0 = X(S - V) - F$$

$$F = X(S - V)$$

$$\boxed{X = \frac{F}{S - V}} \Rightarrow \text{B.E point}$$

Section (2):

1-C 2-d 3-b 4-b

5-b 6-C

marital ⇒ married/engaged.

1-X 2-✓ 3-X 4-✓ 5-X

6-X

SUBJECT: _____ DATE: / /

① Fixed cost = 350 , variable/unit = 8 , S = 15 , X = 200
profit ?? breakeven ??

profit = Revenue - exp
= SX - F - VX
= (15 × 200) - 350 - (8 × 200) = 1650

$$\text{breakeven} = \frac{F}{S-V} = \frac{350}{15-8} = 50 \text{ units}$$

② "A" F = 65000, V = 10, S = 18

$$B.ep = \frac{F}{S-V}$$

$$\frac{65000}{18-10} = 8125 \text{ unit}$$

"B" F = 34000, V = 14, S = 18

$$B.ep = \frac{F}{S-V}$$

$$= \frac{34000}{18-14} = 8500 \text{ unit}$$

③ V/unit = 20 S = 50 F = 150 p if X = 60 ?? Bep ??

p = rev. - exp
SX - F - VX
(50 × 60) - 150 - (20 × 60) = 1650

$$Bep = \frac{F}{S-V} = \frac{150}{50-20} = 5$$